UAV-UGV-based System for AoI minimization in IoT Networks

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***Abstract*—Most of the recent Internet of Things (IoT) applications are highly dependent on the freshness of collected data from IoT devices. Therefore, the concept of Unmanned Aerial Vehicle (UAV) assisted IoT has received a lot of interest where UAVs are deployed as data collectors and can effectively meet the requirement of IoT applications in terms of reducing the Age of Information (AoI) metric of collected data. However, as it is widely known, UAVs are energy-constrained devices with limited computational and communication capacities, preventing them from timely completing their missions. This energy issue can be addressed by deploying fixed charging stations, but this option would interrupt UAVs from their missions, and they should regularly return to these stations to charge their batteries. In this paper, a UAV data collector is dispatched to serve a set of IoT devices while being permanently followed and supported by an Unmanned Ground Vehicle (UGV) as a mobile charging station. Considering an unknown dynamic IoT environment, the problem of AoI minimization is reformulated as a Markov Decision Process (MDP). Indeed, we employ the multi-agent deep Q-network (MADQN) method called AGAIN to minimize the average AoI of IoT devices, timely recharge the UAV using UGV, and optimize the UGV trajectory on the ground by considering the terrestrial obstacles and the IoT movements. A series of simulations are run to demonstrate the efficiency of the proposed method in reducing the AoI of IoT devices compared to baseline methods.**

***Index Terms*—Unmanned Aerial Vehicle (UAV), Wireless Power Transfer (WPT), Unmanned Ground Vehicle (UGV), Data Collection, Internet of Things (IoT), Age of Information (AoI).**

I. Introduction

The evolution of IoT systems over the past few years has rapidly advanced to support many elements of our everyday life. With the increasing amount of generated data by Internet of Things (IoT) devices, it has become necessary to timely collect this data and forwards it to a centralized entity for processing and appropriate decision-making. In the current state-of-the-art [1], it has been shown that Unmanned Aerial Vehicles (UAVs) are characterized by their low deployment costs and could be a promising solution for optimally serving IoT devices and performing data collection (DC). Moreover, UAVs could support real-time IoT applications with freshly collected data by minimizing the IoT devices' Age of Information (AoI) metric. However, the use of UAVs has been hampered by a few technical issues, including their limited energy capacity, restricted transmission power to forward data, and the need to continuously update their trajectories for optimally serving IoT devices.

To ensure the proper functionality of real-time IoT applications, UAV data collectors need to continuously gather fresh information. The AoI metric is used to assess the freshness of gathered data, which is the period of time between the time the data was generated at the source and the current time, and its reception by the UAV data collector [2]. In the literature, the AoI minimization problem has attracted significant attention from the research community in various domains, such as vehicular networks, edge computing, and UAV-assisted communications. For instance, Zhang *et al.* [3] proposed a cluster-based IoT with an AoI-based data collection paradigm where the UAV trajectory and hovering points are jointly optimized. In [4], the AoI minimization problem in UAV-aided mobile edge computing networks has been investigated by considering revealed channel access attacks. The authors of [5] designed a UAV-aided AoI-aware data gathering strategy across IoT networks where various metrics are optimized, such as IoT transmission power, hovering locations, bandwidth allocation, and UAV flight speed. As a drawback, most of the earlier studies, including those discussed above, did not consider possible UAV failures in long-term missions because to their low energy capacity. To get around this issue, some works adopted fixed charging stations to charge UAVs' batteries and enable them to effectively carry out their missions. In [6], the UAV trajectory is optimized to collect data from cluster-based IoT devices, where charging stations are massively installed in each cluster to wirelessly power the UAV. To support UAV data collectors with energy, the authors of [7] deployed a set of flying energy sources that also need to charge their batteries by going back to a fixed charging station. However, installing charging stations could cause various issues, such as interrupting UAVs from their missions, reserving a huge deployment budget, and synchronizing of charging schedule.

All the issues mentioned above motivate us to design a new UAV-based DC system for AoI minimization while overcoming the limited energy capacity of the UAV data collector by deploying an Unmanned Ground Vehicle (UGV) as a mobile charging station. In detail, the UAV trajectory is optimized to optimally serve IoT devices, minimize the expected average AoI, and reduce its energy consumption. It is known that the UAV energy capacity cannot satisfy long-term missions; therefore, we also deployed a UGV and optimized its trajectory, so it always has to follow the UAV data collector

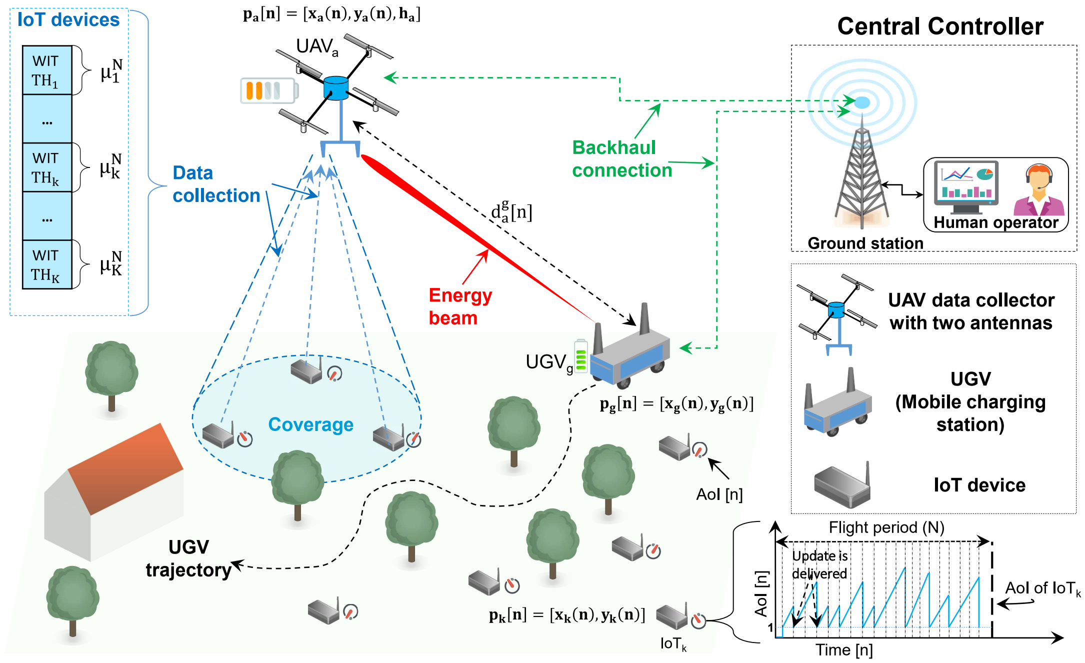


Fig. 1: Motivating scenario.

and appropriately support it with energy. Since the ground environment is supposed to be unknown, dynamic, and full of obstacles, the AoI minimization problem is reformulated as a Markov Decision Process (MDP), and the multi-agent deep Q-network (MADQN) approach is used to reduce the average AoI of collected data, timely charge the UAV using UGV, and optimize the UGV trajectory on the ground by considering the terrestrial obstacles and the IoT movements.

The remainder of this study is outlined as follows. Section II describes on the system concept and formulates the problem statement. Section III provides a brief description of reinforcement learning background and details the adopted MADQN algorithm. The numerical outcomes are covered in Section IV. The paper is finally concluded in Section V.

II. System Model

Consider a UAV-based DC system. As depicted in Fig. 1, the system is composed of one UAV data collector denoted as UAV_a and one UGV denoted as UGV_g playing the role of mobile charging station, which are both controlled by a centralized entity in the form of a base station and human operator. All these elements are deployed to serve a groupe of K dynamic IoT devices represented mathematically by $\mathcal{K} \triangleq \{k = 1, 2, \ldots, K\}$ randomly moving over a specific area not free of obstacles to sense a variety of physical phenomenon. The UAV data collector needs to continuously cover and collect all information about IoT devices, including produced data (*e.g.,* AoI and positions) and report them to the centralized entity for decision-making. The system is analyzed within a certain period T denoted as $t \in [0, T]$. The parameter T is split into N equal time-slots of length δ, *i.e.,* $N = \frac{T}{\delta}$. The selected value of δ is small enough such that at each time slot, UAVs appear to be roughly stationary. The instantaneous position of each ground devices, including UGV is denoted by $p_j[n] = [x_j(n), y_j(n)]^N$ at time-slot n, where $n \in \mathcal{N} \triangleq \{n = 1, 2, \ldots, N\}$, and $j \in \mathcal{K} \cup \{g\}$. The UAV location projected onto the horizontal plane is defined as $p_a[n] = [x_a(n), y_a(n)]^N$. It is important to note that the UAV altitude h_a is fixed at a certain value. The following formula can be used to determine the instantaneous distance between the UAV data collector and any terrestrial devices:

$$d_a^j[n] = \sqrt{\|p_a[n] - p_j[n]\|^2 + h_a^2}, \forall j \in \mathcal{K} \cup \{g\} \quad (1)$$

Table I provides a summary of the key notations used in this work.

TABLE I: List of notations.

Notation	Description
$d_a^j[n]$	Distance between UAV_a and a ground node $j \in \mathcal{K} \cup \{g\}$
$p_i[n]$	Position of node i projected onto the horizontal plane
$E_i[n]$	Residual energy level of node $i \in \{a, g\}$
$E_a^g[n]$	Amount of harvested energy from UGV_g to UAV_a
EC_i	Amount of consumed energy by node $i \in \{a, g\}$
$CP_i[n]$	Activity status of charging process of node $i \in \{a, g\}$
$A_k[n]$	Age of Information (AoI) of IoT_k
$\mathbb{E}[.]$	Expectation by taking into account random IoT mobility
$sp_i[n]$	Instantaneous speed of node $i \in \{a, g\}$
$\Psi_i[n]$	Activity status of node $i \in \{a, g\}$
$r_i[n]$	Reward obtained by node $i \in \{a, g\}$
$s_i[n]$	Environment state observation of node $i \in \{a, g\}$
$a_i[n]$	Action made by node $i \in \{a, g\}$
$Q_i(s_i, a_i)$	Network Q's value at state s_i after conducting action a_i
ε	Exploration coefficient

A. Communication Model

As illustrated in Fig.2, when the UAV needs to recharge its battery, it firstly sends a charging request to the UGV. Then, the UGV transmits the beamforming energy to the UAV.

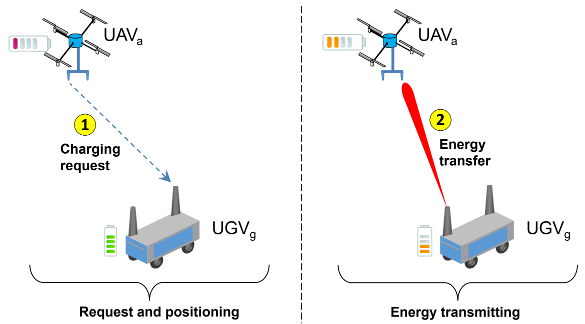


Fig. 2: A UAV-enabled energy harvesting steps.

For this purpose, we suppose that UAV$_a$ has two antennas that operate on orthogonal frequency bands, one for energy harvesting and the other for data transmission to avoid interference. Besides, we suppose that the UGV is outfitted with an antenna dedicated to requests gathering from the UAV as well as another phase-array directional antenna made up of M antenna elements. Each IoT device has a single antenna devoted to data transmission. We represent the Beamforming energy signal by $S \in \mathbb{B}^{M\times 1}$, where $\mathbb{B}^{M\times 1}$ stands for the space of M complex matrices. Therefore, the signal S is derived by referring to the models in [8]:

$$S = \sqrt{P_g}v, \tag{2}$$

where P_g denotes the transmission power of the antenna array. v is the beamforming vector. It is calculated with the help of an estimated channel matrix as $v = \frac{\lambda_1}{\|\lambda_1\|}$, where λ_1 is defined as an eigenvector matching the maximum eigenvalue ϱ of the co-variance matrix $U = uu^H$ and H is the conjugate transpose. In this work, we adopt a normalized vector, *i.e.,* $\|v\|_2^2 = 1$. The harvested energy by the UAV depends on the distance that separates it from the UGV. Hence, according to [9], the harvested energy at time-slot n is given by:

$$E_a^g[n] = \frac{\xi\delta P_g \varrho}{d_a^g[n]} \tag{3}$$

where $0 < \xi \leq 1$ represents the energy conversion efficiency, which should be constant, and δ is the time-slot duration. To get concrete results, we presume that the UAV's maximum energy capacity is set to E_a^{max}. Therefore, the UAV residual energy $E_a[n]$ at time-slot n following UGV energy harvesting is expressed as follows:

$$E_a[n] = \begin{cases} E_a^{max}, & \text{if } E_a[n-1] + E_a^g[n-1] \geq E_a^{max} \\ E_a[n-1] + E_a^g[n-1], & \text{Otherwise} \end{cases} \tag{4}$$

As for the DC process, we adopt a Time Division Multiple Access (TDMA). Indeed, every time-slot n is split into K sub-slots, each of which has a duration of μ_k^n and is dedicated to a given IoT$_k$ under the coverage of UAV$_a$. For the proper functionality of the DC process, we suppose that UAV$_a$ can serve maximum one IoT$_k$ per each sub-slot k. According to [10], if the transmission power of each IoT$_k$ is P_k, where $k \in \mathcal{K}$, the instantaneous rate of IoT$_k$ in a time-slot n is expressed as:

$$Rt_k^a[n] = \mathrm{UI}_k^a[n] B_k^a \log_2\left(1 + \frac{P_k G_k^a[n]}{\sum_{q=1, q\neq k}^{K+g} F_{q,a} + \sigma^2}\right), \tag{5}$$

where $\mathrm{UI}_k^a[n] \in \{0, 1\}$ indicates that IoT$_k$ is connected to UAV$_a$ if $\mathrm{UI}_k^a[n] = 1$, and 0 otherwise. B_k^a and $G_k^a[n]$ denote the allocated bandwidth and the channel gain between IoT$_k$ and UAV$_k$, respectively. $\sum_{q=1,q\neq k}^{K+g} F_{q,a}$ and σ are respectively the overall interference power at UAV$_a$ and the gaussian noise. Let $C_k[n] = \mu_k^n \times Rt_k^a[n]$ be the amount bits transmitted by IoT$_k$ to UAV$_a$ at time-slot n. Therefore, the evolution of the AoI metric $A_k[n]$ for each IoT$_k$ is both illustrated in Fig. 1 and expressed with the following equation:

$$A_k[n] = \begin{cases} 1, & \text{if } C_k[n-1] > C_k^{min} \wedge Rt_k^a[n] > 0 \\ A_k[n-1] + 1, & \text{Otherwise} \end{cases} \tag{6}$$

When an update is successfully received, $A_k[n]$ is reset to 1, otherwise $A_k[n]$ increases by 1.

B. UGV and UAV Energy Consumption

Since the major goal of this work is to allow the UAV energy capacity to meet long-term missions, we solely examine the energy consumption of the UAV and UGV. Therefore, IoT energy consumption is out of the scope of this paper. Furthermore, the UGV is supposed to be deployed in a flat and not inclined area. Thus, the UGV consumes energy at each time-slot n according to the model proposed in [11] as follows:

$$\begin{aligned} EC_g[n] = & sp_g(n)(Res \times Wei + Mass \times acc(n) + IRes) \\ & + b + \epsilon(n), \end{aligned} \tag{7}$$

where $sp_g(n)$ is the UGV's velocity. Res and $IRes$ represent the road rolling resistance and internal resistance coefficients, respectively. Wei and $Mass$ indicate the weight and mass of the UGV, respectively. $acc(n)$ represents the UGV's acceleration, $b(n)$ denotes the consumed energy through onboard equipment, and $\epsilon(n)$ is defined as the model error. It is worth noting that the UGV is also executing the wireless energy transfer to power the UAV using its internal energy. There are two distinct modes: (i) $\alpha_g[n] = 1$ indicates that UGV is undergoing a charging process and (ii) $\alpha_g[n] = 0$ otherwise. Therefore, the UGV energy consumption until time-slot n is given by:

$$Con_g(\{\alpha_g[n], p_g[n]\}, n) = \underbrace{\int_0^n EC_g[n]dn}_{Movement\ energy} + \underbrace{\int_0^n \alpha_g[n] P_g dn}_{energy\ transfer} \tag{8}$$

Regarding the UAV, compared to the energy needed for propulsion, the energy consumed for communication is comparatively minimal. Therefore, we only consider the UAV propulsion energy in this work, which uses the same model as that suggested in [12] and can be calculated as follows:

$$EC_a(sp_a) = \underbrace{BP_b\left(1+\frac{3sp_a^2}{s_{a_{tip}}^2}\right)}_{blade\ profile\ power} + \underbrace{BP_u\left(\sqrt{1+\frac{sp_a^4}{4iv_0^2}}-\frac{sp_a^2}{2iv_0^2}\right)}_{induced\ power} + \underbrace{\frac{1}{2}f_0 adrsHsp_a^3}_{parasite\ power}, \tag{9}$$

where sp_a is the UAV's velocity, $s_{a_{tip}}$ is the rotor blade tip's speed, rs denotes the rotor's solidity, and iv_0 represents the mean induced velocity. BP_u and BP_b denote respectively the induced power and blade profile. The air density, rotor disc area, and fuselage drag ratio are represented by ad, H, and f_0. Consequently, the UAV energy consumption until time-slot n is expressed as follows:

$$Con_a(\{p_a[n]\}, n) = \underbrace{\int_0^n EC_a(\|sp_a[n]\|)dn}_{propulsion\ energy} \tag{10}$$

Consequently, the residual energy of UGV$_g$ and UAV$_a$ at each time-slot n can be calculated as follows:

$$E_a[n+1] = E_a^{max} - Con_a(\{p_a[n]\}, n), \tag{11a}$$
$$E_g[n+1] = E_g^{max} - Con_g(\{\alpha_g[n], p_g[n]\}, n), \tag{11b}$$

C. Problem Formulation

This study aims to optimize the trajectories of drones and UGVs so that the UAV always finds the UGV nearby and is adequately supported by energy to accomplish the entire mission. Furthermore, a set of parameters should also be optimized, such as the UGV energy transfer scheduling, the AoI of all IoT devices, and the energy consumption of both UAV and UGV, all under the dynamics of IoT devices. Thus, we formulate our optimization problem as follows:

$$\max_{p_a[n], p_g[n], \alpha_g[n]} \mathbb{E}\left[\left(\frac{\sum_{n=1}^{N}\sum_{i\in\{a,g\}} E_i[n] \times \sum_{n=1}^{N}\sum_{k\in\mathcal{K}} Rt_k^a[n]}{N \times \sum_{n=1}^{N}\alpha_g[n] \sum_{n=1}^{N}\sum_{k\in\mathcal{K}}\beta_k A_k[n]}\right)\right] \tag{12}$$

$$\begin{aligned}
\text{s.t.}\quad &\textbf{C1:} && \|p_a[n]-p_a[n-1]\|^2 \le (sp_a^{max}\delta)^2,\\
&\textbf{C2:} && \Psi_i[n]\in\{0,1\}, \quad \forall i\in\{a,g\}, \forall n\in\mathcal{N}\\
&\textbf{C3:} && \sum_{i\in\{a,g\}}\Psi_i[n]=2, \quad \forall n\in\mathcal{N}\\
&\textbf{C4:} && \sum_{k\in\mathcal{K}}\mathrm{UI}_k^a[n]\le 1, \quad \forall n\in\mathcal{N}\\
&\textbf{C5:} && d_g^k[n]\ge L, \quad \forall n\in\mathcal{N}, \forall k\in\mathcal{K}\cup\{obstacles\}.
\end{aligned}$$

where β_k is a positive weight reflecting the importance of AoI for different devices. $\mathbb{E}$ is calculated by taking the unpredictability of IoT movement into account. **C1** shows the total distance covered by UAV$_a$ during time interval n. **C2** indicates the current state of activity of both UAV and UGV. **C3** makes sure that the UAV and UGV are fully operational at each time-slot n. **C4** guarantees a safety distance between UGV and IoT devices at each time-slot n to avoid collisions. **C5** ensures that the UGV should always keep a safety distance L from IoT devices and obstacles.

III. Multi-Agent DQN-based Framework: AGAIN

It is noteworthy that (12) is a non-convex mixed-integer optimization problem with no knowledge about IoT mobility. Therefore, an effective strategy is to model the problem (12) as a multi-agent Markov Decision Process (MDP) and adopt the MADQN method to derive a near-optimal solution to it [13]. As illustrated in Fig. 3, the UAV and UGV are regarded as agents transmitting crucial information about the environment and actions to the centralized entity for training. At each time-slot $n \in \mathcal{N}$, each agent $i \in \{a,g\}$ keeps track of the environment's current state $s_i[n]$, performs an action $a_i[n]$ on the basis of a specified policy $\pi_i(s_i[t])$, updates the environment state $s_i[n+1]$, and gets a reward $r_i[n] = \sum_{n=0}^{n}\zeta_n r_i(s_i[n], a_i[n])$, where ζ_n denotes the reward discount factor.

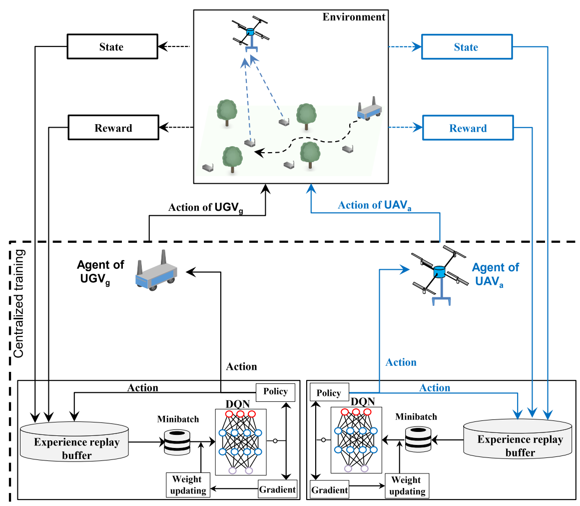


Fig. 3: MADQN framework of AGAIN.

A. State space

The state space $\mathcal{S}$ is considered to be discrete. Therefore the environment's state of each agent comprises the positions of all devices, including UAV and UGV, the energy levels of UGV and UAV, and the availability status of the agent, which is represented as a binary variable $\Psi_i[n]\in\{0,1\}$ (*i.e.,* $\Psi_i[n]=0$ if $(E_i[n]=0)$, and $\Psi_i[n]=1$ otherwise). The environment state's format of each agent $i\in\{a,g\}$ is denoted as $s_i[n]=[p_a[n], p_g[n], p_1[n], \ldots, p_K[n], E_a[n], E_g[n], \Psi_i[n]]$ with a cardinality of $(5+K)$.

B. Action space

The definition of both UGV and UAV action space is $\mathcal{A} = \{\mathcal{A}_{DC}(t), \mathcal{A}_{CP}(t)\}$, where $\mathcal{A}_{DC}(t) = \{\phi_i[t], \Delta_i[t], CP_i[t]\}$ denote the potential actions throughout the DC process. Each element of $\mathcal{A}_{DC}(t)$ is defined as follows:

- $\phi_i[t] \in \{0, \ldots, 2\pi\}$: the horizontal direction of $i \in \{a, g\}$.
- $\Delta_i(t) \in \{0, \ldots, d^{max}\}$: the distance traveled by $i \in \{a, g\}$. When $\Delta_i(t) = 0$ means that node i is staying at the same location, and when $\Delta_i(t) > 0$ means moving.
- $CP_i[t] \in \{0, 1\}$: the charging process status. If $CP_i[t] = 1$, it means that node i is carrying the charging process, and $CP_i[t] = 0$ otherwise.

During the charging process (*i.e.,* $CP_i[t] = 1$), the possible actions of node i are restricted only to $\mathcal{A}_{CP}(t) = \{\phi_i[t], \Delta_i[t], CP_i[t]\}$, where $\phi_i[t] = 0$, $\Delta_i[t] = 0$, and $CP_i[t] = 1$, which means that node i stays at the same location until the charging process is achieved. It is worth noting that the DC process could continue during the charging process since it uses another antenna operating on another orthogonal frequency band. Therefore, the definition of different actions $a_i[n] \in \tilde{\mathcal{A}}(t)$ is expressed as:

$$\tilde{\mathcal{A}}(t) = \begin{cases} \mathcal{A}_{DC}(t), & \text{if } CP_i[t] = 0 \\ \mathcal{A}_{CP}(t), & \text{if } CP_i[t] = 1 \wedge E_a[n] < E_a^{max} \end{cases} \tag{13}$$

C. Reward Function

The main purpose of this work is to optimize the charging process scheduling, serve the maximum number of devices, minimize the distance between UAV$_a$ and UGV$_g$ for increasing the amount of harvested energy, reduce both UGV and UAV energy consumption, and minimize the overall AoI of collected data. Therefore, we reward UAV$_a$ at each time-slot n as follows:

$$r_a[n] = E_a[n] \frac{\sum_{k \in \mathcal{K}} Rt_k^a[n]}{\sum_{k \in \mathcal{K}} \beta_k A_k[n]} - \sum_{n=1}^{n} \alpha_g[n] \tag{14}$$

We also reward UGV$_g$ as follows:

$$r_g[n] = E_g[n] \frac{E_a[n](1 - \Psi_a[n])}{d_g^a[n]} \tag{15}$$

D. Expected penalties

When an action $a_i[n]$ results in collisions with IoT devices or obstacles or crossing the border of the area of interest, a penalty is incurred from the reward of the node i.

E. AGAIN ALgorithm

Each agent $i \in \{a, g\}$ executes AGAIN Algorithm 1 to optimize its actions based on the optimal policy π_i^*. AGAIN is trained over 1000 episodes, each with N steps. It is worth noting that $Q_i^{'}(s_i^{'}, a_i^{'}|\varpi_i^{'})$ and $Q_i(s_i, a_i|\varpi_i)$ denote the target Q value, the approximated Q value, and their respective weights $\varpi_i^{'}$ and ϖ_i. With ε or $1 - \varepsilon$ probabilities, respectively, a random or optimal action $a_i[n]$ is selected. In line 6, the tuples $(s_i[n], a_i[n], r_i[n], s_i[n+1])$ are recorded in the buffer $\mathcal{G}_i$; to prevent temporal correlations. Then, a batch of data is randomly extracted from $\mathcal{G}_i$; to carry out the learning phase (*see* line 7). In line 9, the mean-squared loss between the original value and the updated Q value is minimized to update the value network. Additionally, an update is made every Υ step by setting $\varpi_i^{'} = \varpi_i$ in line 10.

Algorithm 1: AGAIN Algorithm.

1 **begin**
2 **Input:** Initial positions of all devices.
4 **for** $n \leftarrow 1, \ldots, N$ **do**
5 Select an action $a_i[n] = argmax_{a \in \tilde{\mathcal{A}}} Q(s_i[n], a|\varpi_i)$ with a probability $1 - \varepsilon$, otherwise select random actions. Obtain a reward $r_i[n]$ and update to the next state $s_i[n+1]$;
6 Record the transition sample $(s_i[n], a_i[n], r_i[n], s_i[n+1])$ into the buffer $\mathcal{G}_i$;
7 $(s_i[j], a_i[j], R_i[j], s_i[j+1])$ transition samples from a random minibatch of size J;
8 Update weights ϖ_i of $Q_i(.)$:
9 $Loss(\eta_i) =$
$$\mathbb{E}\left[\left(r_i + \zeta \max_{a_i^{'} \in \tilde{\mathcal{A}}} Q_i^{'}(s_i^{'}, a_i^{'}|\varpi_i^{'}) - Q_i(s_i, a_i|\varpi_i)\right)^2\right]$$
10 **Every** Υ **steps:** $\varpi_i^{'} = \varpi_i$
11 **Output:** The optimal policy π_i^*.

IV. Results and Analysis

We assess the performance of AGAIN by conducting a set of simulations. We consider a 2×2 km^2 square region with 20 dynamic IoT devices to perform the training phase of AGAIN over 1000 episodes. We compare the performance of AGAIN with those of Double DQN [14] and DQN methods that are based on a single agent.

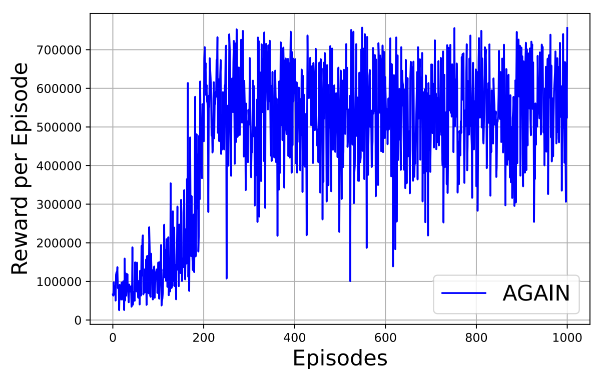
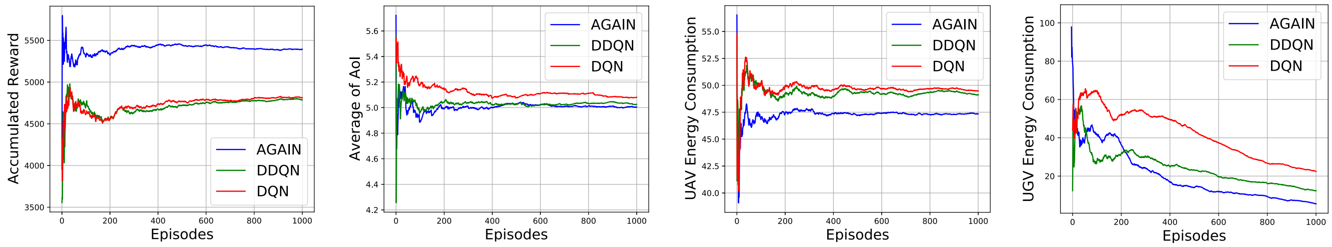


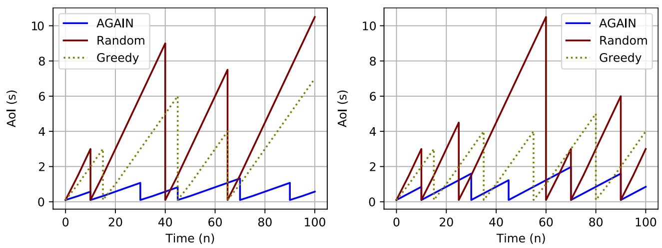
Fig. 4: Reward per episode of AGAIN framework.

As seen in Fig. 4, the obtained rewards using AGAIN converge to peak values after 200 episodes. This is explained by the AGAIN efficient learning of the IoT devices' mobility while taking the appropriate actions regarding the DC process, charging schedule, and energy consumption of both UGV and UAV. Indeed, the accumulated rewards and the UAV residual energy tend to converge over episodes (*i.e.,* Figs. 5(a) and 5(c)). Moreover, we distinguish that AGAIN outperforms DDQN and DQN because of the mitigation of the additional layers used by DDQN and the overestimation in DQN. In Fig. 5(b), the average AoI converge to its lowest value during the whole mission. Moreover, in Fig. 5(d), we also noticed that the UGV residual energy level had been stabilized to 25%. These are explained by the built optimal policy that optimized the UAV trajectories.

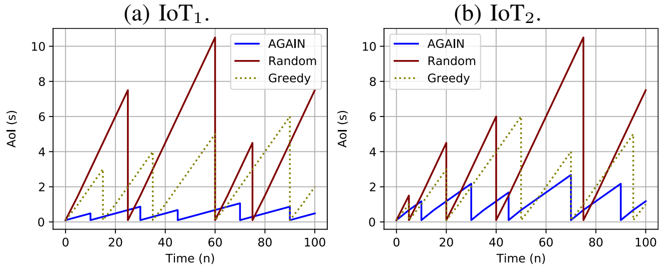


(a) Accumulated reward. (b) Average AoI. (c) Average (%) UAV energy. (d) Average (%) UGV energy.

Fig. 5: Obtained results (Density of IoT devices=20).



(a) IoT_1. (b) IoT_2.



(c) IoT_3. (d) IoT_4.

Fig. 6: Four IoT devices' AoI evolution.

As illustrated in Fig. 6, we calculate the AoI of four IoT devices during the testing phase, and we compare the performance of AGAIN against those of Random and Greedy schemes. The random scheme generates random actions, whereas the second follows a greedy algorithm. All Figures demonstrate another time the effectiveness of AGAIN compared to baseline schemes, which is due to the fact that AGAIN always selects the best strategy to be adapted to the devices' dynamics. Nevertheless, the baseline schemes completely ignore the mobility of devices, resulting in a significant flight time and, thus, higher obtained AoI.

V. Conclusion

This paper studied a UAV-UGV-based system for minimizing the AoI in IoT networks. We deployed a UAV and UGV to synchronize themselves to adequately serve IoT devices and get around the UAV's limited battery capacity. We leveraged an MADQN strategy to optimize the actions of both UAV and UGV for efficiently charging the UAV, minimizing the average AoI, and significantly minimizing the UAV and UGV energy consumption. The generated results have clearly demonstrated these objectives compared to other schemes in which both UAV and UGV fail to be synchronized towards the primary objectives. In the future, we plan to study the continuous state-action space of both UAV and UGV while considering some accurate physical parameters.

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